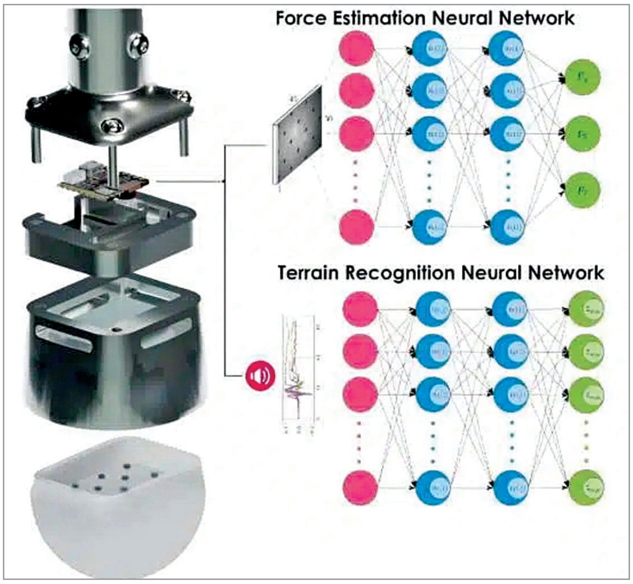
A sensing paw that could improve the ability of legged robots

Researchers from NTNU and IIT Bombay have developed an innovative artificial paw for quadruped robots, enhancing their ability to navigate various terrains. This artificial paw, TRACEPaw, collects sensory data from the environment,



Left image shows the CAD design in an exploded view with the Silicon paw with the pattern on the bottom, aluminium housing in the centre, and the board with camera and microphone. The right image illustrates the neural networks used to estimate force based on the pattern deformation and the neural network used to classify the terrain (Credit: Vangen et al)

processes it with a computer vision model, and predicts terrain characteristics and contact force by assessing surface deformation and soil noise. The system is cost-effective and utilises readily available electronic components. Preliminary experiments in the lab show promising results, indicating that TRACEPaw can significantly improve legged robot agility and adaptability to different terrains. This technology holds potential for real-world applications like search and rescue missions and exploration tasks. The team plans to further improve the system's capabilities in force estimation and soil classification through additional data training.

Advancing hydrogen production with new electrolyser technology

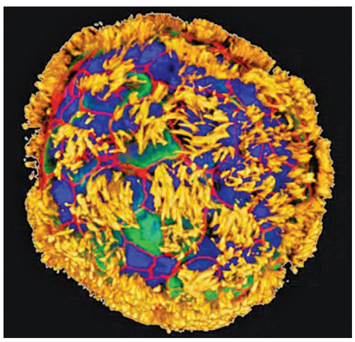
Efficient hydrogen production is crucial for sustainable energy solutions. Researchers at UNIST's School of Chemical Engineering have introduced an innovative ionomer-free technique called m-CCM (catalyst-coated membranes) that integrates a catalyst layer directly between the anion exchange membrane (AEM) and gas diffusion layer, eliminating the need for anion exchange ionomers. The resulting anion exchange membrane water electrolyser (AEMWE) with a platinum group metal-free anode catalyst outperforms MEA-based electrolysers. It achieves an industrial-grade current density of 1A cm^{-2} at 1.79V cell and remains durable for over 200 hours of continuous operation at 50°C in 1M KOH electrolyte.

Advanced imaging technique for multiple cellular signals

MIT scientists have developed an advanced cell imaging method that simultaneously tracks up to seven molecules within living cells. This technique utilises green and red fluorescent molecules that blink at different rates, enabling the observation of target protein levels over time through image capture and computational analysis. By engineering various switchable fluorophores, each blinking at a different rate, and using linear unmixing similar to sound frequency analysis, they can pinpoint the location and timing of each labelled molecule in the cell. Importantly, this imaging can be conducted using a standard light microscope, making it widely accessible for various biological research applications, from studying cell responses to stimuli to exploring growth, ageing, and cancer.

Tiny biobots pioneering neuron regrowth and tissue healing

Researchers from Tufts University and the Wyss Institute at Harvard University have created Anthrobots using human tracheal cells. These tiny robots can move on surfaces and potentially stimulate neuron growth in damaged areas. The Anthrobots come in various shapes and move differently, offering versatility for potential applications inside the body or in lab-grown tissues. They naturally biodegrade in about 45-60 days. In lab experiments, these Anthrobots significantly promoted neural regrowth, suggesting their potential for healing neural tissue. They could have applications in clearing arterial plaque, repairing nerve damage, detecting harmful cells, and targeted drug delivery, ultimately aiding tissue healing and regeneration.



Human tracheal skin cells self-assemble into multi-cellular, moving organoids called Anthrobots. These images show Anthrobots with cilia on their surface (yellow) distributed in different patterns. Surface patterns of cilia are correlated with different movement patterns: circular, wiggling, long curves or straight lines (Credit: Gizem Gumuskaya, Tufts University)